

PHASE IV Final Forensic Case Report

A. Executive Summary (Non-Technical)

A suspicious ZIP file was discovered on a system without a clear origin. The file contained multiple items, including text files, images, logs, and a hidden file. During the investigation, the archive was extracted and analyzed using Kali Linux tools.

The analysis identified a hidden file and a text file containing plain-text credentials. While no active malware was observed in this simulated case, the presence of hidden data and unsecured credentials represents a clear security risk.

Overall, the investigation demonstrates how simple artifacts—such as compressed files and hidden content—can expose sensitive information and create potential entry points for unauthorized access.

B. Incident Overview

- **Scenario:** Mystery ZIP file discovered on system
- **Environment:** Kali Linux (Virtual Machine)
- **Scope:** File-level forensic investigation of compressed archive
- **Objective:** Determine contents, identify hidden/sensitive data, assess risk

The case simulates a common situation where unknown archives are found and must be examined safely and methodically.

C. Evidence Summary

- **Evidence ID:** ZIP-001
- **Type:** ZIP archive (suspicious.zip)
- **Source:** Local VM (Kali Linux)
- **Acquisition Method:** File creation and controlled simulation (Phase 2)
- **Contents Identified:**
 - notes.txt
 - creds.txt
 - image1.jpg
 - log_file.log
 - .secret.txt (hidden)
- **Chain of Custody:** Maintained within a controlled lab environment
- **Hashing:** Not enforced in this simulation; in real cases MD5/SHA256 required

D. Methodology

The investigation followed a structured approach:

1. **Evidence Identification** – Suspicious ZIP file located

2. **Initial Inspection** – Contents listed without modification
3. **Extraction** – Archive unpacked in controlled environment
4. **File Enumeration** – All files (including hidden) identified
5. **Content Analysis** – Files reviewed for sensitive or suspicious data

Tools Used:

- unzip (archive extraction)
- ls -la (file enumeration, including hidden files)
- cat (content inspection)
- strings (optional, for text extraction)

These tools were selected for their reliability, simplicity, and effectiveness within Kali Linux.

E. Detailed Findings

1. Hidden File Detection

A hidden file (.secret.txt) was identified during analysis. Hidden files are often used to conceal data from casual observation.

Finding: Presence of concealed data

Impact: Indicates potential attempt to hide information

2. Plain-Text Credentials

The file creds.txt contained:

```
password=admin123
```

Finding: Credentials stored in plain text

Impact: High risk — could allow unauthorized system access

3. General File Review

Other files (notes.txt, log_file.log, image1.jpg) did not show immediate malicious content but were included in the investigation for completeness.

Finding: No active malware identified

Impact: Low direct threat, but supports context of activity

F. Timeline Reconstruction

Although simulated, the logical sequence of events is:

1. Files created within a directory
2. Directory compressed into ZIP archive

3. ZIP file discovered
4. Archive extracted and analyzed

In real investigations, timestamps from file systems, logs, and memory would provide precise timing.

G. Conclusion and Assessment

The investigation determined that the ZIP file contained both normal and sensitive data. The presence of hidden content and unsecured credentials suggests improper data handling and potential security risks.

While no malware execution was confirmed, the findings demonstrate how even simple artifacts can expose vulnerabilities. In a real-world case, additional steps would include:

- Hash verification of all files
- Full disk or memory analysis
- User activity investigation
- Network traffic review

H. Recommendations

- Avoid storing credentials in plain text
- Scan all unknown archives before opening

- Implement file monitoring and logging
- Enforce user security awareness

I. Appendices

- Command logs (zip, unzip, ls, cat)
- Screenshots of extraction and file analysis
- Evidence file listing
- Tool versions (Kali Linux packages)